Jeong-woo Han

(519) 722 9439
4644 Doug Wright Dr, Burlington ON, L7M 0L2, Canada
fwthree3@gmail.com
https://fwthree3.github.io

CURRENT POSITION

Research Associate, Mechanical and Mechatronics Engineering **7.2026 -Present**
University of Waterloo, Canada
- Developing a robotic HIFU treatment system

WORK EXPERIENCE

Senior Autonomy Engineer **4.2025 -7.2026**
Mirsee Robotics Inc, Canada
- Real-time motion planning and control of dual arms of a humanoid robot
- Custom multi-axis Ethercat driver with URDFs and ROS2
- Framework development for autonomous pick-place operations

Research Associate, Mechanical and Mechatronics Engineering **9.2022 -4.2025**
University of Waterloo, Canada
- Manufacturing-AI project manager
- Planning and control interfaces of a 6DOF parallel robotic manipulator
- High-level controller for a mobile manipulator

Lead Researcher, Medical Robotics Institute **7.2012 -8.2017**
Hyundai Heavy Industries, Republic of Korea
- Gait rehabilitation robot, MORNING WALK, from scratch to commercialization (Certificated Korean FDA /Launched the developed robot in public hospitals)
- 3-year of robot H/W development and 2-year of S/W enhancement
- Promoted early to Lead Researcher based significant contribution to the robot development

EDUCATION

Ph.D. in Mechanical and Mechatronics Engineering **9.2017 -8.2022**
University of Waterloo, Canada
- Dissertation: Safe and Efficient Navigation of a Mobile Robot: Path Planning Based on Hierarchical Topology Map and Motion Planning with Pedestrian Behavior Model

M.Sc. in Mechanical Engineering **3.2010 -8.2012**
Yonsei University, Seoul, Republic of Korea
- Thesis: Design and Hardware Validation using CoP of Walking Assistive Lower Limb Exoskeleton for Paraplegia Patient

B.Sc. in Mechanical Engineering **3.2003 -2.2010**
Yonsei University, Seoul, Republic of Korea

SKILLS

Engineering Abilities
- Robot motion planning and control of arm-manipulation / mobile base with fluency in ROS/ROS2
- Design and system integration of robotics systems, including sensor implementations and embedded system development
- Structural, kinematics, and dynamics simulation and analysis

Experienced Techniques
- PCB design, ANSYS, Recurdyn

Key Techniques
- ROS/ROS2 ★★★★★
- C language ★★★★★
- Python ★★★★★
- MATLAB ★★★★★
- Solidworks ★★★★★
- Embedded ★★★★☆

LEADERSHIP

- **Group representative of 230 new employees**, *Hyundai Heavy Industries* **7.2012**
- **Group representative of 80 military trainees,** *Republic of Korea Air Force* **6.2005**

PUBLICATIONS / PATENTS

Papers

- Lari, Salman, **Jeong-woo Han**, and H. J. Kwon. "Evaluation of Internal Flaw Geometric Characteristics Using Artificial Neural Networks." *Proceedings of the 20th World Conference on Non-Destructive Testing* (2024).
- Sadeghi-Goughari, Moslem, Hossein Rajabzadeh, **Jeong-woo Han**, and Hyock-Ju Kwon. "Artificial intelligence-assisted ultrasound-guided focused ultrasound therapy: a feasibility study." *International Journal of Hyperthermia* 40, no. 1 (2023): 2260127.
- **Han, Jeong-woo**, Soo Jeon, and Hyock Ju Kwon. "Hierarchical Topology Map with Explicit Corridor for global path planning of mobile robots." *Intelligent Service Robotics* 16, no. 2 (2023): 195-212.
- Cenerini, Joseph, Mohamed W. Mehrez, **Jeong-woo Han**, Soo Jeon, and William Melek. "Model Predictive Path Following Control without terminal constraints for holonomic mobile robots." *Control Engineering Practice* 132 (2023): 105406.
- Kim, Jung-Hoon, **Jeong Woo Han**, Deog Young Kim, and Yoon Su Baek. "Design of a walking assistance lower limb exoskeleton for paraplegic patients and hardware validation using CoP." International Journal of Advanced Robotic Systems 10, no. 2 (2013): 113.
- **Han, Jeong-woo**, Soo Jeon, and Hyock-Ju Kwon. "A new global path planning strategy for mobile robots using hierarchical topology map and safety-aware navigation speed." In *2019 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pp. 1586-1591. IEEE, 2019.
- **Han, Jeong-woo**, and Young Hwan Kim. "Torque Compensation Mechanism for End-Effector Type Gait Rehabilitation Robot." *Annual Conference of the Korean Society of Mechanical Engineers* 15 (2015): 714-718.
- Kwon, Suncheol **Jeong-woo Han**, Jin Hyuck Choi, and Young Hwan Kim. "Development of End-effector Type Gait Rehabilitation Robot with Body-weight Supporting Saddle." *Technical Conference on Rehabilitation Engineering and Assistive Technology Society* 11 (2014): 28-29.
- **Han, Jeong-woo**, Il Hwa Hong, Heung Soon Lim, and Sung Hyun Jung. "Development on a Needle Insertion End-effector for Intervention Surgery." *Annual Conference of the Korean Society of Mechanical Engineers* 3 (2013): 30-31.
- **Han, Jeong-woo**, Yoon Su. Baek, Jung-Hoon Kim. "Design of Exoskeletal Assistance Robot for Walking and Rehabilitation." *Annual Conference of the Korean Society for Precision Engineering* 10 (2011): 851-852.

Patents

- Walking training apparatus, (US)10894182, (KR)102601919 (2016, granted in 2021)
- A Seating-type Robot for Gait Trainer Apparatus, (US)10881575, (EP)3299001B1, (CN)107708641, (JP)06720218, (KR)101623686 (2015, granted in 2020-2021)

- Seating-Type Gait Rehabilitation Apparatus having Lateral Driving Function, (KR)102474762 (2015, granted in 2022)
- Foothold of Robot Improved Function of Load Cell, (KR)102311565 (2015, granted in 2021)
- Body Weight Support part of rotation type, (KR)102484973 (2015, granted in 2023)
- Apparatus for Supporting Walking Trainee and Apparatus for Training Walk Comprising the same, (KR)102296449 (2014, granted in 2021)
- Foothold of Robot for Rehabilitation of Walking, (KR)102261918 (2014, granted in 2021)
- Apparatus for Transporting Trainee and Apparatus for Training Walk Comprising the Same, (KR)102296444 (2014, granted in 2021)
- Apparatus for Supporting Trainee and Apparatus for Training Walk Comprising the Same, (KR)102263758 (2014, granted in 2021)
- Seating-Type Supporting Unit and Seating-Type Apparatus for Training Walk using The Same, (KR)102098959 (2013, granted in 2020)
- End-effector for Intervention of Inserting Needle, (KR)102158499 (2013, granted in 2020)
- Drive device using driving source of only one, (KR)102034517 (2013, granted in 2019)
- Medical Robot Driving Safety for Multiple Safety Apparatus, (KR)101826576 (2012, granted in 2018)